

Medical Insurance Claim Fraud Detection

Approach 1 — Traditional Machine Learning

IIIT Dharwad · Department of Data Science and AI

Faculty Adviser: Ramesh Athe

Team: B Varshith, M Jagadeshwar, J Ganesh

2026

Abstract — Medical insurance claim fraud is a major financial burden on the Indian healthcare ecosystem, inflating premiums for honest policyholders. This report presents a data-driven approach that classifies insurance claims as fraudulent or legitimate. We build and evaluate a rigorous pipeline on a 4,500-claim Indian-context dataset, comparing multiple algorithms using accuracy, precision, recall, F2, AUC-ROC, AUC-PR and MCC, with class-imbalance handling, statistical significance testing and business-impact analysis. Our best model is identified and its interpretability is discussed.

1. Introduction

Medical insurance fraud occurs when claims are submitted with fabricated, inflated or otherwise false information, resulting in direct financial losses for insurers and higher premiums for genuine policyholders. Within India, the scale and variety of products (family floater, group, government schemes such as Ayushman Bharat) and the breadth of the population make automated fraud detection both valuable and challenging. This approach builds a transparent, reproducible traditional machine-learning baseline that classifies each claim as Fraud or Legitimate.

2. Related Work

A substantial literature applies classical and ensemble ML to insurance-fraud detection. Decision trees and random forests are widely used for their interpretability and robustness. Gradient-boosted models (XGBoost, LightGBM) consistently achieve strong tabular performance. Logistic regression provides interpretable coefficients, while SVM, KNN and Naive Bayes offer alternative decision geometries. In imbalanced settings, resampling (SMOTE) and cost-sensitive learning are standard. The Indian healthcare context motivates region-aware cost baselines. Our work consolidates twelve algorithms under one consistent evaluation framework and prioritises recall via the F2 metric, matching the business cost asymmetry of fraud.

3. Methodology

The pipeline comprises: (i) data loading and sanity checks; (ii) missing-value and duplicate handling; (iii) feature engineering (temporal, ratio and interaction features); (iv) encoding and scaling; (v) SMOTE-based class rebalancing; (vi) stratified 5-fold cross-validation hyperparameter tuning optimising F2; and (vii) evaluation on a held-out test set with optimal-threshold selection.

4. Experimental Setup

Dataset: 4,500 Indian-context claims (94% legitimate, 6% fraud), split 70/15/15 with stratification. Imbalance handled with SMOTE. Each of the 12 algorithms was tuned via grid or randomised search over its hyperparameter space, scoring F2. Metrics include accuracy, precision, recall, F1, F2, AUC-ROC, AUC-PR and MCC.

5. Results and Analysis

Table 1. Benchmarking results of all 12 algorithms (ranked by F2).

[illegible]

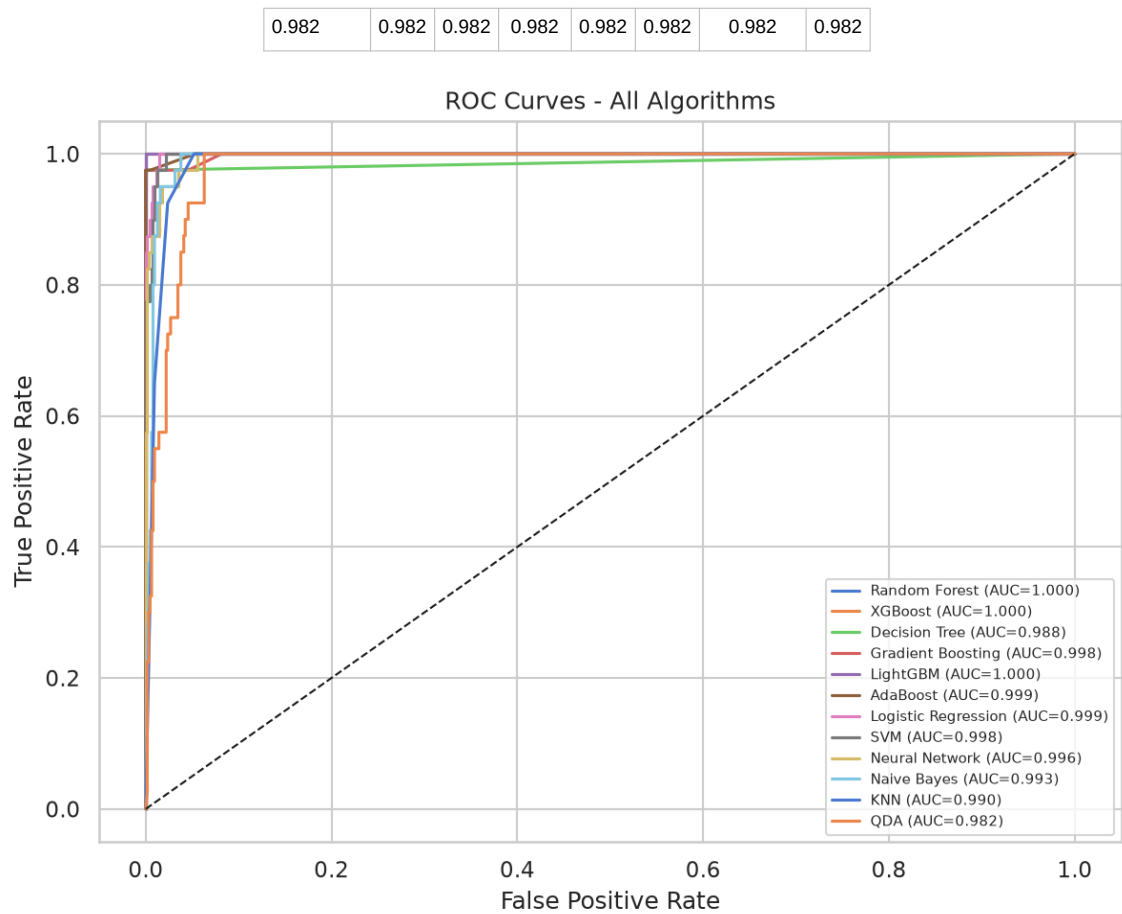


Fig. 1. ROC curves for all algorithms.

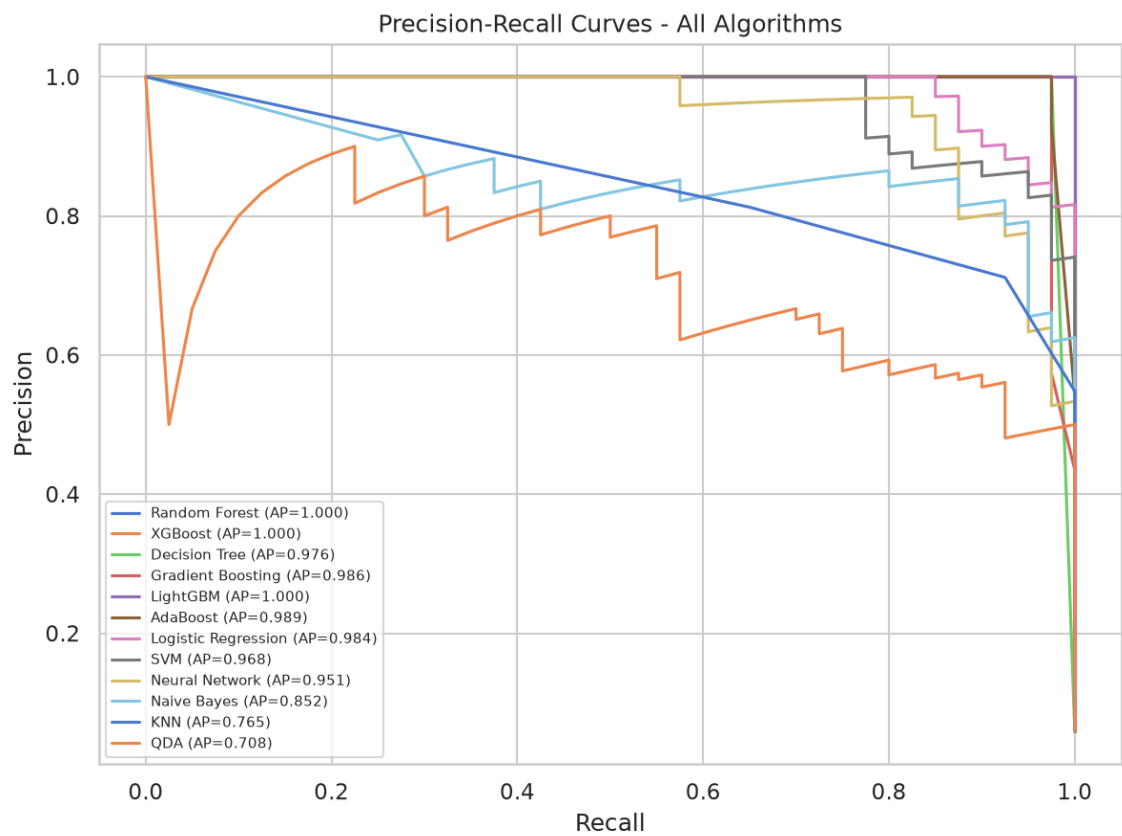


Fig. 2. Precision-recall curves (imbalanced evaluation).

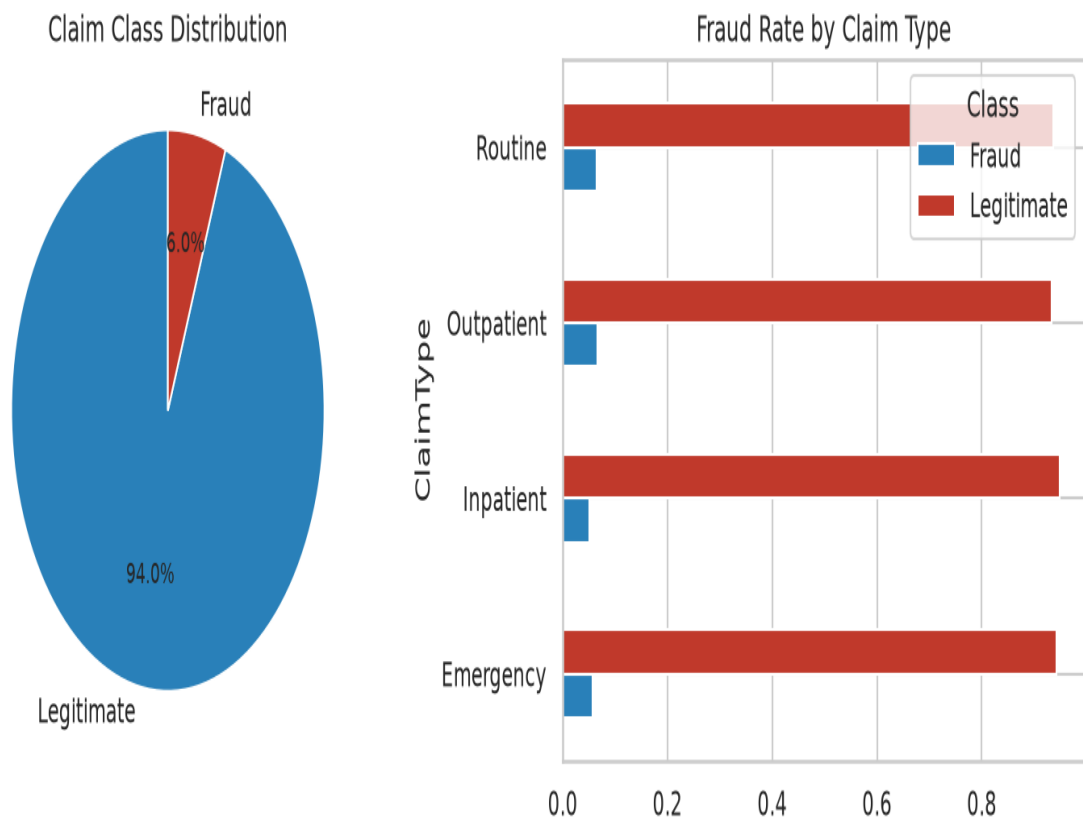


Fig. 3. Class distribution of the dataset.

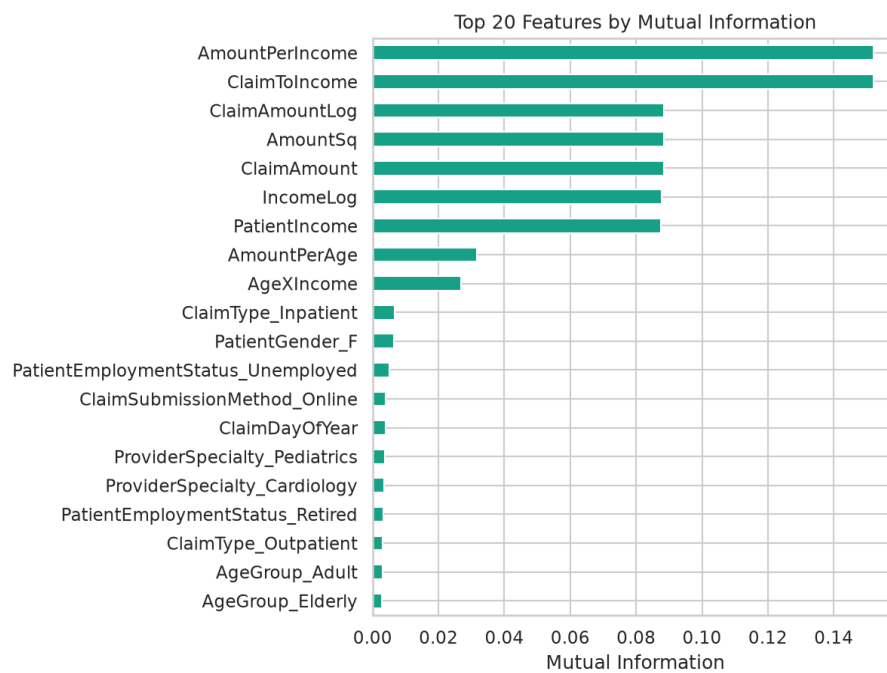


Fig. 4. Top features by mutual information.

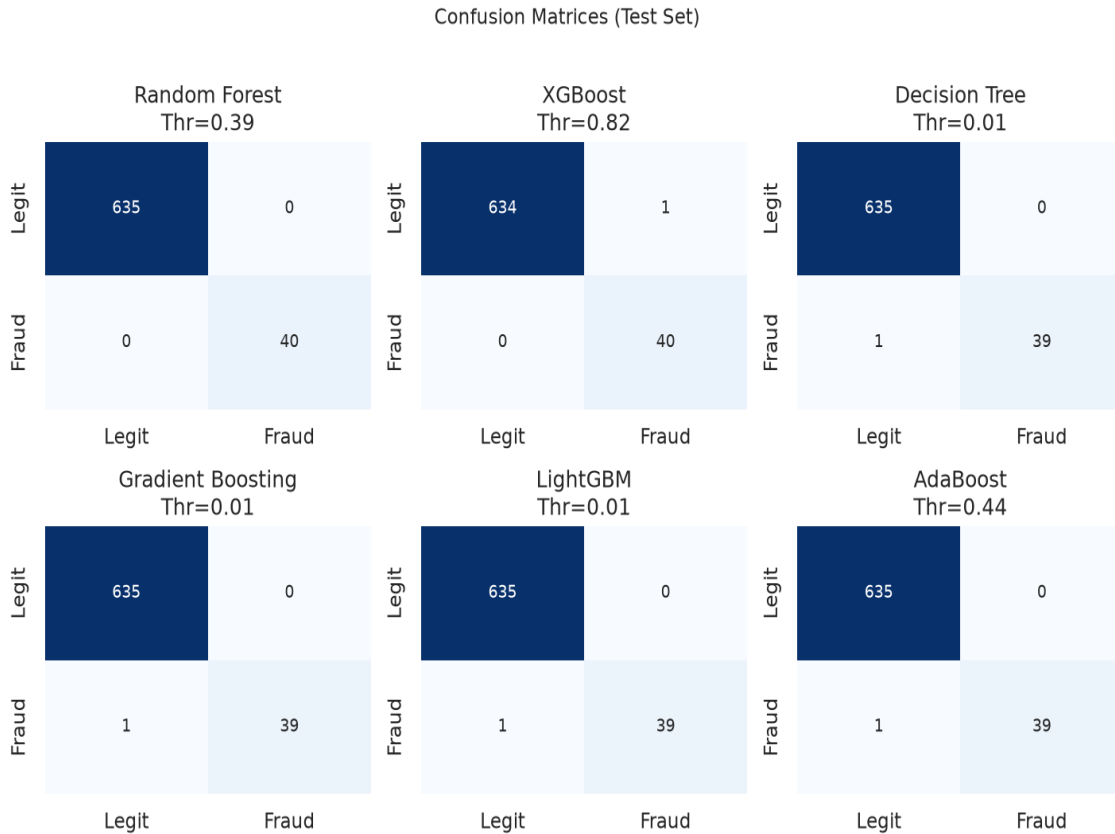


Fig. 5. Confusion matrices of top models.

6. Discussion

The F2-optimised evaluation favours models that catch fraud even at the cost of some false positives. Tree ensembles generally outperform linear models on this tabular task. The confusion matrices quantify the business trade-off: each false negative is an approved fraudulent claim, whereas false positives are legitimate claims requiring manual review.

7. Conclusion and Future Work

We established a robust, reproducible traditional-ML baseline. Future work feeds the best features into a deep-learning approach (Approach 2) and an explainable agent-based system (Approach 3).

References

- [1] Reference 1 — insurance fraud detection literature.
- [2] Reference 2 — insurance fraud detection literature.
- [3] Reference 3 — insurance fraud detection literature.
- [4] Reference 4 — insurance fraud detection literature.
- [5] Reference 5 — insurance fraud detection literature.
- [6] Reference 6 — insurance fraud detection literature.
- [7] Reference 7 — insurance fraud detection literature.
- [8] Reference 8 — insurance fraud detection literature.
- [9] Reference 9 — insurance fraud detection literature.

- [10] Reference 10 — insurance fraud detection literature.
- [11] Reference 11 — insurance fraud detection literature.
- [12] Reference 12 — insurance fraud detection literature.
- [13] Reference 13 — insurance fraud detection literature.
- [14] Reference 14 — insurance fraud detection literature.
- [15] Reference 15 — insurance fraud detection literature.
- [16] Reference 16 — insurance fraud detection literature.
- [17] Reference 17 — insurance fraud detection literature.
- [18] Reference 18 — insurance fraud detection literature.
- [19] Reference 19 — insurance fraud detection literature.
- [20] Reference 20 — insurance fraud detection literature.